

MD RAHIDUL ISLAM

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About Me

A Computer Science and Engineering student and aspiring full-stack developer, hardworking, detail-oriented, and passionate about technology, with a strong team-player mindset built through academic and team projects. Hands-on experience building web applications using Next.js, NestJS, TypeScript, and PostgreSQL — including responsive interfaces, REST APIs, authentication, and role-based access control — consistently delivered by working closely within collaborative teams to meet deadlines. I value clean, maintainable code and practical problem-solving, and I'm seeking an entry-level software development role where I can contribute to production-quality products while growing alongside an experienced engineering team.

Educational Background

Bachelor of Science (BSc), *American International University-Bangladesh*

Department: Computer Science and Engineering

Dhaka, Bangladesh

Expected: December 2026

Higher Secondary Certificate (HSC), *Notre Dame College, Mymensingh*

Department: Science

Mymensingh, Bangladesh

2021

Secondary School Certificate (SSC), *Poncahashi High School*

Department: Science

Mymensingh, Bangladesh

2019

Projects

ChargeHubEV | *Next.js, NestJS, TypeScript, PostgreSQL, Tailwind CSS* | 🐙 [Backend](#) | 🐙 [Frontend](#)

- Engineered a full-stack EV charging platform with a responsive Next.js frontend and a modular, maintainable NestJS backend.
- Designed RESTful APIs for user accounts, station bookings, payments, authentication, and technician workflows using TypeORM and PostgreSQL.
- Secured application workflows with JWT authentication, bcrypt password hashing, role-based access control, request validation, and reusable form components.

Smart Metro E-Ticket Management System | *PHP, JavaScript, AJAX, MySQL, HTML, CSS* | 🐙 [Source Code](#)

- Developed a database-driven metro e-ticketing platform supporting passenger accounts, route-based fare calculation, ticket booking, and validation.
- Implemented asynchronous AJAX workflows with PHP and MySQL to update booking data dynamically and reduce unnecessary page reloads.

Java Tour Management System | *Java, Object-Oriented Programming* | 🐙 [Source Code](#)

- Designed a Java-based tour management application for organizing travel packages, customer information, reservations, and payment records.
- Applied encapsulation, inheritance, abstraction, and modular class design to separate business logic and improve code maintainability.

Smart Coffee Shop Management System | *C#, .NET Framework, MySQL*

- Developed a C# desktop application for order processing, automated billing, inventory tracking, and customer record management.
- Designed the MySQL relational schema and integrated it with a modular .NET interface for reliable real-time data operations.

Additional Projects

Smart Railway Station Visualization | *C++, OpenGL, GLUT* | 🐙 [Source Code](#)

Course: *Computer Graphics, AIUB*

- Created an animated railway-station simulation with trains, tracks, platforms, and environmental elements.
- Applied geometric transformations, event handling, rendering, and timer-based animation using OpenGL and GLUT.
- Organized visual components into reusable drawing functions to improve scene consistency and simplify animation updates.

Smart Health Monitoring System | *ESP32, Sensors, IoT, Blynk*

Course: *Microprocessor and Embedded Systems, AIUB*

- Built an ESP32-based system to collect sensor readings and transmit live health data wirelessly.
- Integrated embedded hardware with an IoT dashboard for remote monitoring and device-status updates.
- Combined real-time sensing, wireless communication, and status visualization in a practical embedded monitoring workflow.

Skills

Computer & Technical Skills:

- **Programming Languages:** Python, JavaScript, TypeScript, C++, C#, Java, PHP, HTML, CSS.
- **Frameworks & Libraries:** Node.js, NestJS, Next.js, React.js, TypeORM, Material-UI, DaisyUI, Tailwind CSS, .NET, GUNA UI, Swing, OpenGL.
- **Databases & Tools:** PostgreSQL, MySQL, Oracle SQL, Git, GitHub, Postman, VS Code, Visual Studio, Linux (Ubuntu/WSL), Blynk, Figma, Trello, Canva, Cisco Packet Tracer, Multisim, AutoCAD, Microsoft 365.
- **Core Concepts:** Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Computer Networking, Operating Systems, REST API Development, Authentication and Authorization.

Language Skills:

- Bangla: Native
- English: Professional working proficiency
- Hindi: Conversational

Research Experience

Industrial Defect Localization with Explainable Heatmaps

 [Repository](#)

| Focus: Computer Vision, Explainable AI | Tech: Python, PyTorch, TensorFlow, OpenCV

- Conducted a comparative study of Grad-CAM and patch-based heatmap methods for lightweight, explainable industrial defect localization using the MVTec AD dataset.
- Developed an end-to-end pipeline for image preprocessing, CNN inference, heatmap generation, and localization analysis using Python, PyTorch, TensorFlow, and OpenCV.
- Evaluated visual explanations across multiple defect categories to compare localization clarity, model interpretability, and prediction reliability for practical quality-inspection scenarios.

Adversarially Robust Network Intrusion Detection

 [Repository](#)

| Focus: Machine Learning, Cybersecurity | Tech: Python, Random Forest, XGBoost, Neural Networks

- Designed a multi-model network intrusion-detection pipeline on the NSL-KDD dataset by combining Random Forest, XGBoost, and neural network approaches within a multi-agent framework.
- Prepared network-traffic data through categorical encoding, feature scaling, label mapping, and structured train-test evaluation to support consistent model comparison.
- Tuned decision thresholds and evaluated accuracy, precision, recall, and F1-score to analyze detection reliability and classification trade-offs across normal and malicious traffic.

Achievements

CISCO IT Essentials: PC Hardware and Software | [Certificate](#)

11 Jun 2023

Two-time 1st Prize, Inter-School Parade Competition

2017–2018

Led the Poncahashi High School parade team to first place twice.

1st Place, School-Level Scholarship Examination

2017–2018

Extracurricular Activities

General Member, Notre Dame Science Club

2019–2020

- Assisted with the National Science Festival and participated in workshops, exhibitions, and academic seminars.

Parade Team Leader, Poncahashi High School

2017–2018

- Led practice sessions, coordinated members, and maintained discipline during national and school events.

Hobbies

- Traveling, Reading books,Playing guitar,Watching movies